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21. *Poučení z krizového vývoje ve straně a společnosti po XIII. sjezdu KSČ: Rezoluce k aktuálním otázkám jednoty strany* (Prague, 1971), p. 22.
22. *The Restoration of Order: The Normalization of Czechoslovakia*, trans. A. G. Brain (London, 1984), p. 17.
23. For more details see, "Normalizace v Národním divadle," *Listy*, 1, 1971, no. 1, pp. 17-19.
24. John Barton, *Playing Shakespeare* (London, 1984), pp. 191-2.
25. For Havel's reaction to these rumors see, "Fakta o představení," pp. 173-4 and 179.
26. For more details see, "Horní Počernice—všecko vystoupit," *Listy*, 6, 1976, no. 2, pp. 26-7; and "Společenská rubrika," *ibid.*, p. 48.
27. "Toward a Theory of Schizophrenia," quoted from G. Bateson, *Steps to an Ecology of Mind* (New York, 1972), pp. 217-18.
28. "Moc bezmocných," quoted from *O lidskou identitu*, p. 60.
29. *Ibid.*, p. 61.
30. *Ibid.*, p. 63.
31. "Dopis Gustávu Husákovi," quoted from *ibid.*, p. 24.
32. "Moc bezmocných," p. 70.
33. "Dopis Gustávu Husákovi," p. 23.
34. For more details about this case see, "První a poslední role 'kapitána' Minaříka," *Listy*, 6, 1976, no. 2, pp. 7-8; and "Skandál v Římě," *ibid.*, no. 4, p. 11.
35. *Žebrácká opera: Na téma Johna Gaye*, quoted from V. Havel, *Hry 1970-1976: Z ústraní, Audience, Vernisáž, Ztížená možnost soustředění, Žebrácká opera, Asanace, Protokol, Horský hotel, Spiklenci, Mstitel, Slavnostní proslov, Zahradní slavnost* (Toronto, 1977), p. 120; pagination for all the subsequent quotations from *Žebrácká opera* are marked directly in the body of my essay. For their English translations I am indebted to Paul Wilson whose kind help I am pleased to acknowledge.
36. "A Theory of Play and Fantasy," quoted from *Steps to an Ecology of Mind*, p. 183.
37. "Začarovaný kruh," *Divadlo*, 18, January, 1967, p. 3.

SOUND PATTERNS IN THE *SLOVO O POLKU IGOREVE*: COMPARATIVE AND THEORETICAL PERSPECTIVES ON OLD RUSSIAN POETICS

Olga Yokoyama and Brent Vine

In the library of literature concerning the poetics of the Russian epic tale *Slovo o polku Igoreve*, surprisingly little work has been done with respect to its sound texture; and with very few exceptions, most published work concerns the rhythmic or metrical structure of the poem.[1] This statistical stylistic study provides basic data for a partial analysis of the sound texture of the *Slovo*. In particular, we are concerned with the distribution of consonants in the piece as a whole and in its various parts. On the basis of these data we examine underlying patterns of consonant distribution in the *Slovo*, and their function in the poetic organization of the tale. These findings will lead us to comment on some of the theoretical and methodological implications of the study for the analysis of the sound structure of poetic texts in general, focusing on the question of so-called "sound symbolism," and on alliteration. Finally, we will suggest that the nature of alliteration in the *Slovo*, as compared with alliteration in other Old Russian texts and in other Indo-European poetic traditions, has important consequences for the notorious problem of the authenticity of the *Slovo*.

1. Methodology

1.1 *Statistical Procedure.*

This study is based on the following tabulations and calculations.[2] First, the number of occurrences of each consonant (N_b, N_p, N_v, N_f, N_m, . . .) in the entire text of the *Slovo* was counted. Then, the average frequency of each consonant was determined by dividing the number of occurrences of each consonant (N) by the total number of consonants (ΣN). Consonants were also grouped into phonetic classes (liquids, sonorants, affricates, velars, etc.), and the average distribution of each phonetic class per total number of consonants was calculated in the same way (for example, $N_{liq}/\Sigma N$, $N_{son}/\Sigma N$, etc.).

After establishing, in this way, the average occurrence of each consonant in the text as a whole, the same calculations were performed on five individual passages. Thus, by first determining the total number of occurrences of each consonant (and phonetic class) in a given passage (n_b, n_p, n_v, . . . ;

n_{liq}, n_{son}, . . .) as well as the total number of consonants in the passage (Σn), the average frequency of each consonant (or phonetic class) in the passage could be determined in the same way as for the *Slovo* as a whole ($n_b/\Sigma n$, etc.; $n_{liq}/\Sigma n$, etc.).

It may be helpful, before proceeding, to offer a brief example of these calculations. Consider the relative frequency of the lateral *l*. The *Slovo* as a whole contains 703 *l*'s and 7,293 consonants altogether. By dividing these two ($N_l/\Sigma N$), we arrive at a frequency of 9.64 per 100 consonants. If, however, we examine "Jaroslavna's Lament" (versets 168-183), we find 56 *l*'s in this passage, as compared with 403 consonants altogether; dividing these two figures ($n_l/\Sigma n$) yields a relative frequency of 13.90 *l*'s in the "Lament" (i.e. 13.90 *l*'s per 100 consonants)—a frequency that is evidently higher than the frequency of 9.64 for the *Slovo* as a whole.

What remains is to arrive at an index of deviations such as this, i.e. deviations of the average frequency per consonant (or phonetic class) in a given passage ($n/\Sigma n$) from the average frequency of the same consonant (or phonetic class) in the text as a whole ($N/\Sigma N$).[3] Although this could in principle be done in various ways, the most straightforward method of arriving at such an index, for our purposes, was simply to divide the smaller of $n/\Sigma n$ and $N/\Sigma N$ by the larger. The closer this number is to 100%, the closer the frequency of a given consonant (or class) in a particular passage matches the frequency of that consonant (or class) in the text as a whole. To continue with the example already used above: if the average frequency of *l* in the *Slovo* as a whole is 9.64%, and the average frequency of *l* in the "Lament" is 13.90%, then the frequency of *l* in the *Slovo* as a whole deviates from the average frequency of *l* in the "Lament" by 69.4 (i.e. 9.64/13.90). The number 69.4, in other words, signifies that the frequency of *l* in the "Lament" deviates from (and in this case exceeds) the overall average of *l*'s by more than 30%. As a working hypothesis, moreover, we will in general consider only deviations greater than 30% to be "significant."[4]

In an illuminating discussion of statistics and the analysis of literary style, Richard Bailey (1971) proposed a method of determining the "prominence index" of particular phonemes in a poetic text, as compared with their overall frequency in the language, in an attempt to quantify "phonemic foregrounding" in poetic texts. Despite the evident similarity between our concerns and those of Bailey, it is important to observe the differences between the two approaches. Given the non-technical nature of this presentation, we comment here not on the details of statistical method, but on certain basic theoretical assumptions.

As a way of gauging Bailey's approach, consider the following statement (1971:31): "Positive rank-differences would seem to have greater perceptual value than negative ones. That is, readers will hardly be aware of the relative absence of a particular phoneme, but greater exploitation of a phoneme than might normally be expected will usually be noticed." Such a statement shows, however, that Bailey is concerned largely with quantifying only those phonetic features that are perceptually prominent in relatively gross ways—indeed, in ways determined *a priori* by the analyst. Our approach, on the other hand, is more narrowly descriptive: both kinds of deviations—i.e. relatively high and relatively low frequencies—are given equal attention. Mathematically speaking, after all, a higher than average occurrence of some element(s) in a sample entails a lower than average occurrence of some other element(s) in the same sample; there is, then, no justification for ignoring the distribution of low-frequency elements in a text that is marked for the high frequency of some sound(s).

This descriptive orientation, in turn, means that we have made no assumptions about which sorts of phonetic behavior might have "perceptual prominence", or more generally whether the quantification of what we perceive as "prominent" should be the exclusive goal of statistical stylistics. Moreover, given the importance of subliminal patterning in poetic texts (see Jakobson 1970), a consideration nowhere addressed by Bailey, it has become clear that the phonetic organization of poetic texts may involve structures of considerable subtlety. Ideally speaking, judgments about perceptually prominent features in poetic texts—subliminal or otherwise—should be supported by psychological testing; but in the absence of such studies, there is, then, no *a priori* reason to exclude from consideration any particular form of phonetic patterning. In the case of low frequency sounds or classes of sounds, we will suggest in §3.3 below, in the context of our discussion of alliteration, that despite Bailey's claim to the contrary, this sort of patterning may in fact be significant in certain ways, and deserves to be explored further.

1.2 *On the Relative Frequency of Consonants in the Slovo.*

Before turning to certain textual preliminaries and to the analysis of the data, it is important to explore in somewhat more detail one of the fundamental concepts involved in this study, namely the notion of the average (relative) frequency of consonants in a text.

To our knowledge, there exists no previous calculation of the average frequency of consonants in Old Russian texts. How can we know, then,

that the patterns uncovered in this study are "typical" or "normal" for Old Russian? This is an important point: if we are going to examine, precisely, *deviations* from some norm (as might occur, namely, in particular passages), we must have some way of determining that the ranges described as "normal" are really normal; lacking this, there would be no way of evaluating how "significant" alleged deviations from this claimed norm would be. In a strict statistical sense, such a problem is only partly real, since any deviation from a proposed "norm" can in theory be evaluated with respect to chance. But in the present case there is no need to resort to complex tests of this sort, since previous studies of the relative frequencies of consonants in other languages provide confirmation of the norms arrived at here.

On the basis of relatively limited data from English and Czech, Bailey tentatively suggested (1971:33-4) that "the distribution of phonemes in languages operates within highly restrictive limits," and that "the rather small variation between languages suggests that the frequency series occupied by phonemes is relatively fixed." He concluded, then, that "[t]he approximate similarity of redundancy values thus suggests that a linguistic universal constrains human speech in the deployment of phonemes." There was no need, however, for this conclusion to have been phrased so tentatively. Building largely on pioneering work by Zipf, Pierre Guiraud—one of the *maîtres* of statistical stylistics—had published already in 1959 a study of the relative frequency of phonemes (see Guiraud 1959, ch. IX: "Structure diacritique et structure statistique des systèmes phonologiques"). Guiraud's analysis of nearly twenty languages (including several non-Indo-European ones) led him to confirm and to refine a number of the statistical norms Zipf had already arrived at for the distribution of phonemes in phonological systems.[5] For Russian itself, a remarkable and little-known study is due to Peškovskij, in which he tabulated the absolute and relative frequencies of all of the sounds occurring in a text consisting of 10,000 graphemes (1925). Peškovskij himself, in his lucid concluding remarks (1925:181), saw quite clearly that the overall patterns he had uncovered were likely to be universal within certain limits, and even suggested the potential interest of deviations from these norms that might be associated with differences in literary genres.

The lack of statistics concerning the relative frequency of phonemes in Old Russian, then, is of little consequence for verifying the degree to which our *Slovo* tabulations reflect a norm, since normal ranges for phoneme distribution can be calculated on a more or less universal basis (at least for the European languages sampled by Zipf and Guiraud). Using Peškovskij's tabulations as a check against Guiraud's figures for 17 European languages (including Russian, and based on a count of 10,000 phonemes), Figure 1

shows at a glance that our counts of voiced and voiceless stops in the *Slovo* (based on a comparable total figure of 7,293 consonants) present precisely the same patterning of relative frequency with regard to voicing (and nearly the same patterning with regard to place of articulation) as compared with Peškovskij's and Guiraud's figures:

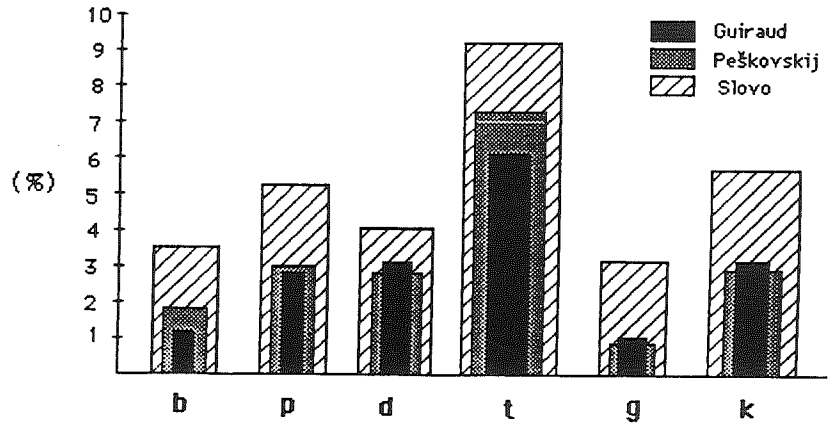


Figure 1: Relative Frequency of Stops

For present purposes, it would be otiose to present similar verifications for other consonants and consonant classes. We can be reasonably confident, in general, that our total tabulations for the consonants of the *Slovo* correspond to established norms, thereby providing a basis for our calculations of deviation.[6]

We stated above that for the purposes of this study, only deviations of 30% or more will be considered "significant." But given what has already been said about both subliminal patterning and the need for psychological testing, it should be clear that this figure is partly arbitrary. We are reasonably certain, however, that from a statistical point of view and from an artistic point of view, deviations greater than 30% are not likely to be due to "chance," i.e. to the effects of random distribution of consonants within the language. The precise interpretation of these deviations, not all of which are due to the poet's conscious efforts, can only be imperfectly and partially broached in a preliminary study such as this, and remain open to much more thorough investigation.

1.3 *Textual Preliminaries.*

Certain textual issues have a direct bearing on the methodology of this study; foremost among them, of course, is the problem of the text itself.

The textual history of the *Slovo* is well known, and need not be recapitulated here. Suffice it to say that in the absence of an authoritative manuscript tradition, it was necessary to choose one of the several "reconstructed" versions of the text, so as to provide a consistent body of data as a basis for our tabulations. Jakobson's text was chosen for this purpose, largely because of its ready accessibility, and also because of the explicit nature of Jakobson's textual assumptions.[7] To be sure, statistical data based on other reconstructions would vary somewhat, depending on the editors' treatment of a number of scribal, orthographic and dialectal problems posed by the text. It is unlikely, however, that the element of variation introduced by the occasional different readings of other reconstructions would significantly affect the conclusions reached here.[8]

The other general difficulty concerns the partly indeterminate phonetic value of the phonemes of the *Slovo*. For consonants, voicing assimilation and palatalization constitute pervasive and systematic problems, since the extent to which yers had fallen at the time of the *Slovo*'s composition cannot be safely determined.[9] We have avoided the problem of yers, and more specifically their effect on voicing assimilation in new consonant clusters, by generally disregarding the voiced/voiceless opposition in the present treatment of consonants. The discussion will operate, then, with reference to classes like "velars," "stops," "fricatives" etc., although we will occasionally point to situations in which voicing needs to be mentioned. As for palatalization, we have assumed, as a working hypothesis, that it was not a relevant factor for consonantal texture at the time of the composition of the *Slovo*, and the tabulations were accordingly performed without reference to this opposition.[10]

Other indeterminacies—involving, for example, tabulation of the (partly subphonemic) glide [j] in prevocalic and intervocalic positions, or the classification of graphic (i.e. transcribed) <*v*> (fricative or sonorant?)—have no bearing on the data selections presented below; we therefore reserve comment on such issues for a fuller presentation of the data.

2. Survey of the Data

2.1 *Svjatoslav's Dream* (93-99).[11]

The distribution of consonants in this passage is highly marked in several respects. In terms of classes, the frequency of nasals is significantly high (with a deviation index of 66.8). This is a striking feature, as our data indicate that the distribution of nasals rarely varies from the average. The passage is also markedly low in liquids (69.1)[12], especially *l*'s (56.2). In terms of individual consonants, the most striking feature is perhaps the high frequency of *x* (36.8).

2.2 *Jaroslavna's Lament* (168-183).

The most marked feature of the "Lament" is the high frequency of affricates (63.4). Note that the frequency of liquids is higher than average (78.9), but apparently not significantly; we return to this point in more detail below. Stops and fricatives are moderately lower than average (stops 86.1, fricatives 76.1), but among fricatives, the frequencies of *x* (37.6), *ž* (54.4) and *š* (53.8) are extremely low.

If we consider the component parts of the "Lament" separately, however, an important fact emerges: the distribution of consonants in Jaroslavna's direct speech and in the narrator's four interposed introductions does not mirror the overall distribution of consonants in the "Lament" as a whole, as this was just outlined. It appears, rather, that Jaroslavna's words show a "milder" picture of the same tendencies found in the "Lament" as a whole, whereas the narrator's four introductions (168, 172, 177, 181) actually exhibit an intensification of the features just mentioned. Thus, in Jaroslavna's direct speech, affricates are less drastically marked than in the "Lament" as a whole, as are liquids. Nor are stops and fricatives quite as infrequent as in the whole of the "Lament." In the narrator's introductions, however, the degree to which both affricates and liquids exceed the average frequency is extremely high. Most other consonants (except nasals, which are average) show a significantly lower than average frequency. In addition to the low frequency of *x*'s, other velars are also of low occurrence, as are *ž* and *š*. Note also *d*, which is here extremely low, but in Jaroslavna's speech (and the passage as a whole) about average. These relationships emerge clearly from the following sample figures, in which capitalized terms indicate higher-than-average frequency, and lower-case terms lower-than-average frequency:

	"Lament"	*Jaroslavna*	*Narrator*
AFFR	63.4	71.8	47.7
LIQ	78.9	86.6	62.9
stops	86.1	90.7	67.3
frics	76.1	78.0	70.7
x	37.6	50.4	(0)
ž	54.4	73.6	(0)
š	53.8	54.9	51.6
d	86.1	91.9 (D)	22.0

These relationships among the parts of the "Lament" suggest that the narrator's interaction with the lament proper functions in an extremely subtle but important way, as far as sound texture is concerned. It is actually the narrator's words—and not Jaroslavna's—that set the "key" for the entire passage in the most marked way, with respect to certain features, such as the high frequency of *č* (40.7, but only 71.1 in Jaroslavna's direct speech). These salient features actually become somewhat attenuated by the lament itself, though in the passage as a whole they nonetheless retain their saliency (e.g. *č*: 59.5 for the "Lament" as a whole). In addition, the narrator's words introduce other "marked" effects not found in Jaroslavna's speech at all (such as the extremely low frequency of *d*). We will consider some further implications of this passage below, together with the notion of "keyed" passages in general.

2.3 *The Chase* (200-208).

Despite the relative brevity of this passage (240 consonants in all), nearly all consonants—and thereby nearly all consonant classes—approximate the normal distribution of consonants in the *Slovo* as a whole fairly closely.[13] Nevertheless, this humorous conversation between the two Polovcians deviates from the average in a single way, which we can now easily quantify as follows: velars are strikingly more frequent than average (65.3), and among them, the frequency of *k* is remarkably high (44.6). Given what has already been said about the difficulty of pressing voicing distinctions, one should be hesitant in distinguishing between the behavior of *g* and *k*, especially since *Kъza* may actually have been pronounced *Gza*, as the editio princeps has it. But there can be no doubt that this orchestration of *k*'s was deliberate, as is indicated by the abundant use here of the preposition *kъ*, which appears no less than three times in this passage (203, 205, 207). The full significance of this fact emerges if we examine the usage of this preposition in the rest of the *Slovo*: it seems clear that *kъ* was optional, to judge from its use in 9, as contrasted with its omission three other times (in 19, 77 and 100) in similar contexts.

Finally, note that the high frequency of velars is accompanied by just one other marked feature, namely a remarkably high frequency of affricates as compared with their overall average (54.5), and here the salient affricate is *c* (29.7).

2.4 *The Battle* (37-56).

What distinguishes this passage on the whole is the absence of any marked class of consonants with respect to relative frequency. The passage approximates, then, the average distribution of consonants.[14] It is, in fact, the longest section of those surveyed (687 consonants), and one might wonder whether this effect is simply a function of sample size. We postpone consideration of this point briefly until examining the next section.

2.5 *The Exodus* (184-191).

This passage resembles the preceding one with respect to the absence of a marked class of consonants.[15] Among individual consonants, the frequency of *x* is extremely low (0%), as is that of *š* (54.3); but these consonants are rare in any case (1.33% and 1.84% in the *Slovo* as a whole, respectively), and the low numbers involved render the percentages in this passage not wholly reliable. Perhaps more significant is that *g* is extremely high (52.5); but on the other hand, *k* is only moderately high (78.7). This brings velars to a deviation index of only 75.8, which seems to indicate that the frequency of velars as a class is not significant in this passage.

It seems clear, then, that the near-average distribution of consonants in some passages is not a function of the size of the sample, since no correlation holds between near-average consonant distribution and sample size. Thus, among the five passages analyzed in this study, those unmarked for any "key" contain 299 ("Exodus") and 687 ("Battle") consonants, while those marked for particular keys contain 166 ("Dream"), 240 ("Chase"), and 403 ("Lament") consonants.[16]

3. Discussion

3.1 *Patterns of Marking in Individual Passages.*

Some passages in the *Slovo* are marked for certain consonants or classes of consonants; in a general way, we have said that a significantly high or low frequency of a given class or sound sets the "key" for an entire passage. Other passages, in contrast, have no overall key, but exhibit a near average distribution of all consonants. This is not really a new observation; what we are suggesting, rather, is a method of quantifying (and ultimately assessing) part of what is often vaguely and subjectively referred to as the "tone" or "tonality" of a poetic text, familiar both from Slavic and general poetics. It is important to note, however, that traditional discussions of "tonality" tend to focus on what is often called the "orchestration" of vowels; but there is no justification for this restriction. In passages such as the "Dream," the "Lament," or the "Chase," as discussed above, the "tonality" of these passages is as much a factor of consonantal as vocalic patterning (if not more so, in some cases).

With regard to the "orchestration" of consonantal texture, the data point

to certain further patternings. The role of affricates is particularly interesting. A high frequency of affricates (or, in the case of the "Dream," a relatively high frequency of the affricate *č*) can be observed in all three of the keyed passages. In the "Dream," *č*'s accompany a high frequency of nasals; in the "Lament," they accompany (a) a high frequency of liquids (especially in the narrator's introductions), and (b) low frequencies of various items, as seen in the chart in §2.2 above; and in the "Chase" (here mostly *c*) they accompany the rather prominent velars. In the passages without a particular key, on the other hand, affricates are less prominent. We hypothesize, then, that affricates constitute a powerful instrument in the *Slovo* for intensifying already marked sound textures. They do not pattern with *particular* classes, but rather with *any* class that serves to determine the key of a given passage. This apparently automatic association between affricates and keyed passages amounts to a sort of implicational relationship, in the Jakobsonian sense, and it would not be surprising if further research revealed the existence of similar relationships involving other sounds and/or keyed patterns.

A final note on the behavior of affricates concerns *šč*. Given uncertainties about the phonetic value of *šč* at the time of the compostition of the *Slovo*, this symbol was tabulated as a fricative, and not an affricate. Note, however, that *šč* seems to pattern much like the affricates *č* and *c*, as just described; if *šč*, then, is counted as an affricate, it would clearly render the affricate behavior just sketched even more dramatic. Thus in the "Dream"—a marked passage full of interesting phonetic effects—*č* is relatively frequent, but its deviation index of 73.5 does not quite bring it into the region of significance, as we defined it for the purposes of this study. And since *c* does not occur at all, the combined index for these two affricates is 96.9, which gives the impression that affricates appear in the passage with about normal frequency. But *šč*, which is otherwise extremely infrequent, occurs twice in this relatively short passage, yielding a deviation index of 46.7 (i.e. significantly more frequent than normal). Both of the instances of *šč*, moreover, appear in prominently alliterated or paronomastic parts of the passage:

96. Sypaxutь *mi tъščimi tuly* poganyxъ *Tъl*kovinъ
98. *V*ъsju noščь *sъ večera busovi vrani vъzgra*jaxu

Although there is a danger of circularity, to be sure, it is not excluded that a sufficiently detailed and controlled theory of the poetic behavior of affricates in the *Slovo* could provide an argument for the phonetic nature of *šč*, if the behavior of *šč* turns out to be as systematic as it appears to us at present.

3.2 *On "Sound Symbolism."*

These considerations regarding the patterning of sound classes have a direct bearing on the theory of sound symbolism in poetics, and in particular its widespread use as a tool in the analysis of poetic texts. Here, as is well-known, there are essentially two traditions: first, a marked tendency among literary critics to assess the "effects" of particular sounds in poetic texts in terms of a host of vague and subjective notions[17]; and second, a more scientific approach that seeks to isolate, via controlled studies (and partly through psychological testing), the precise iconic associations of particular sounds.[18] In either case, however, the objective is the same—namely the identification of more or less precise referents with particular sounds.

The present analysis suggests certain limitations on the evaluation of sounds. Consider, for example, the evaluation of affricates in literary texts. In a rich, monograph-length study, Patruno Bonora (1966-67) has investigated the history and function of the affricate *č* in German, and his exhaustive survey of the synchronic behavior of *č* in German vocabulary leads him to conclude as follows (1966-67:459) "Il *č*. . .ha sempre, in tutte le parole, in qualsiasi posizione, valore fonosimbolico. . . .Tutte le parole que contengono questa parola, o hanno mutato la loro forma primitiva per assumere il *č*, per motivi espressivi, o sono creazioni nuove. Tutte attestano que *č* è un fonema comune, usuale, familiare, e dimostrano inequivocabilmente il valore espressivo di questo fonema."

In typical fashion, Patruno Bonora is not careful to distinguish his claims about the "expressive" nature of *č* in German from similar (and widely repeated) claims about *č* in more universal terms. In the *Slovo*, however, the funcion of affricates, as we have seen, must be described differently: in general, affricates are used to reinforce the effect of another prominent class in a phonetically marked passage. Therefore, the concomitant and indiscriminate attraction of affricates to a variety of consonants and consonant classes renders it meaningless—indeed, impossible—to assign a fixed symbolic or expressive value to affricates as a class (much less to particular affricates). On the other hand, it is not excluded that at least some sounds or classes of sounds are associated with a particular symbolism; and it may well be that some inherently "expressive" quality of affricates is ultimately responsible for their association with phonetically marked classes in the poetic technique that underlies the *Slovo*. But the symbolic values of particular sounds are notoriously difficult to prove, beyond certain more or less obvious (and usually iconic) associations for a small number of sounds[19]; and as for affricates, the precise nature of any universal "expressive"

associations such sounds may have remains quite undefined. Instead, we suggest that the notion of sound symbolism should be reevaluated (or enhanced) so as to include purely *functional* phenomena, like the "implicational" behavior of affricates in the *Slovo*—entirely without recourse to statements about the putative symbolic, psychological, emotional (etc.) content of particular sounds. For certain sounds and in the phonetic technique of certain poetic traditions, we would claim, such statements are of a secondary nature (if not altogether spurious), as compared with the actual functioning of particular consonants and classes with respect to such features as frequency distribution.

3.3 *On Alliteration.*

Another relationship can be demonstrated between the "key" of a passage and the orchestration of other consonants within it: namely, alliterations in a keyed passage tend to be composed of infrequently occurring sounds. In the "Dream," for example, there are two obvious alliterated groups—one based on *t*, and one on *b*:

(96) *t*ъščimi *t*uly. . .*t*ъlkovinъ
(99) *b*oloni *b*ěšja d*ь*br*ь*sky

Despite the increase in frequency one might assume to be a natural consequence of alliterated sequences, both sounds occur with lower than average frequency in the passage. The same relation holds for alliterated *p* in the "Chase"; again, *p* occurs with low frequency in the passage, despite the following prominent sequence:

(208) *p*očьnuть. . .*p*ъticě biti vъ *p*oli *P*olovьčьskomь[20]

Alliterations that are foregrounded in this way, i.e. those based on low-frequency consonants in a given passage, natually have a high degree of perceptual prominence, while the key stratum of a passage is partially, or perhaps largely, subliminal. As a result, it may be that the lower the relative frequency of a sound, in a given passage, used in an overtly alliterated phrase, the more striking the effect of this alliteration against the background of a particular key. This subtle organization of alliterative technique in the *Slovo*, then, shows that Bailey's insistence on the insignificance of low-frequency phonemes for poetic structures (see §1.1 above) is misconceived. On the contrary, the facts just discussed suggest that the effects of low-frequency distribution of phonemes in poetic texts deserve to be explored in much greater detail.

The importance of alliteration in the *Slovo*, as well as its relationship with alliterated patterns in Old Russian prose saga literature, are by now well known, thanks to Čiževsky's widely cited study of 1949, building in

part on earlier work by Ržiga.[21] While Čiževsky's descriptive observations about alliteration in the *Slovo* and in Old Russian prose are accurate and insightful, his discussion of the origin of alliteration in Old Russian and of alliterative traditions elsewhere in Indo-European (1949:129-30) is superficial and misleading in certain important respects.

After noting that alliteration is a favorite poetic device of some (but not all) Indo-European literatures, Čiževsky observes, quite correctly, that the most famous Indo-European manifestation of alliteration—the so-called "Stabreim" of Germanic literatures—"has almost no resemblance to Old Russian alliteration," since it tends to be linked with a specific poetic rhythm and involves fairly precise restrictions on the number of alliterated words in a poetic unit (such as the verse line).[22] Čiževsky then surveys (in a cursory and partly inaccurate way[23]) several other Indo-European alliterative traditions, and concludes as follows: "The use of alliteration seems to be linked with the peculiarities of prosodic structure: it occurs prevailingly in languages with accent on the initial syllable of words or roots. As the Old Russian language did not offer such a prosodic feature, the possibility of a borrowing of this device, probably from Germanic poetry, is not excluded."

Čiževsky's conclusion, however, is a *non sequitur*. Borrowing would only be an attractive explanation if the alliterative system of Old Russian was structurally similar to the Germanic types, despite the fundamental difference in the Slavic and Germanic accentual systems. But by Čiževsky's own admission, as already noted, this is simply not the case. It is important to specify what Čiževsky's unsatisfactory survey of Indo-European alliterative traditions does not make clear, namely that languages with initial stress tend to develop not just a propensity for alliteration, but alliterative systems with a high degree of structure, as in Germanic. This is true not only of Old Irish[24], among the languages Čiževsky mentions, but also of Latin; here, however, Čiževsky's reference to medieval Latin is less apposite than Archaic Latin, where a prehistoric initial stress (whose effects are visible in the numerous vowel reductions of Classical Latin) was intimately bound up with the pervasive alliteration of Archaic Latin poetry.[25] Čiževsky's statement that "Greek literature. . .has almost no alliteration" is by and large correct; but here again, an important exception appears in the archaic lyric poetry composed in the Lesbian dialect of Aeolic Greek (known to us primarily through the poetry of Sappho and Alcaeus), where the recessive accent characteristic of this dialect was surely a significant factor in the development of the alliterative tradition in which the Lesbian poets operated. The more episodic alliteration typical of the *Slovo*, however, is clearly of a different type: it is hardly the major constitutive feature of Old Russian

epic verse, which is not surprising, in view of the prosodic system of Old Russian. Rather, alliteration contributes an important element—but only one element—of a complex system of phonetic repetitions and phonetic highlighting in general, as emerges clearly from many of Jakobson's well-known discussions of Old Russian sound texture in terms of "paronomasia," to which we will return below. Given, then, fundamental differences between Old Russian and Old Germanic poetry not only in prosody but in the alliterative systems themselves, the hypothesis of borrowing from Germanic must be considered an unattractive one.

A simpler and more satisfying explanation is at hand. More recent studies of ancient Indo-European poetics (see especially Schmitt 1967 and 1968; Gamkrelidze and Ivanov 1984:832-42) have led to important reconstructions of Indo-European poetic practice. Perhaps the best-known studies of this sort involve the reconstruction of Indo-European metrical structures. Less well-known, but of comparable importance, are discoveries concerning certain "genres" of Indo-European poetic composition and the poetic techniques associated with them, beyond purely metrical features. Among the most important of these "genres" is the so-called "alliterative *carmen*" of Indo-European,[26] best known from its reflexes in Vedic poetry and in Archaic Latin prose. The salient features of this form of composition have been discussed in some detail by Schmitt (1967:206-7, with further references), and can be summarized briefly as follows: it is a highly rhythmic style that is neither prose nor verse; there is a striking tendency toward symmetry, often involving two or more parallel cola (with two-member parallels being especially characteristic, often with a high degree of synonymy between them); the parallel structures are often linked via alliteration.

Recall that the form of alliteration in the *Slovo* to which Čiževsky drew attention was precisely that which "binds the final words of short syntactic units" (1949:125)—a pattern so striking and pervasive in the *Slovo* as not to require repeating Čiževsky's numerous examples, which could easily, as he points out, be multiplied. It is, however, worth repeating Čiževsky's important finding of the pervasiveness of this alliterating style in Old Russian chronicle literature, and especially its restriction to those sections that retell the old sagas; Čiževsky concludes his survey of alliteration in Old Russian prose, then, by insisting that "particular attention should be paid to the epic character of the majority of those sections in the Chronicles in which we come upon alliteration. It is even more remarkable that there is almost a complete lack of alliteration in the ancient sermon literature which includes quite a few works with an extraordinarily developed literary style that employs the most refined literary devices and a somewhat rhythmic

language which is also one of the stylistic peculiarities of the epic parts in the Chronicles" (1949:129). Thus with respect to patterns of alliteration (and also with respect to a certain rhythmic character, the more precise analysis of which is beyond the scope of this study), the *Slovo* finds its closest stylistic analogue in the epic prose of saga literature, i.e. the prose genre most directly linked with the secular (and ultimately pagan) tradition. It is simplest to suppose, then, that by and large, these salient features of native Old Russian epic poetics are not the result of Germanic influence, but rather continue certain traditional features of Indo-European poetic technique, as seen particularly in the Indo-European alliterative *carmen* style.

3.4 *On the Authenticity of the Slovo.*

In the "Retrospect" to volume IV of his *Selected Writings*, Roman Jakobson commented that "The sound texture of the *Slovo* . . . still waits for a systematic inquiry" (1966:682), and provisionally offered numerous detailed analyses of the intricate and effective organization of sounds in the *Slovo*.[27] In one of these (1966:683-4), Jakobson studies the third section of the *Slovo*, "with its integral, sumptuous tissue of rapturous assonant words . . . and anagrammatic allusions." In comparing this section with corresponding verses of the *Zadonščina*, however, Jakobson finds that "The *Zadonščina* shatters this whole complex and harmonious edifice into dissociated debris." Jakobson's important 1972 study of parts of the *Slovo* reached similar conclusions: as part of his comparative study of the parallel passages *Slovo* 40-41 and *Zadonščina* 28-9, Jakobson notes: "We should mention in passing that the triple assonance of initial vowels characteristic of the *Tale* . . . has come to naught in the *Zadonščina*" (1972:329). In the same study, note further Jakobson's comparisons (1972:330) of the *Slovo*'s "considerable emphasis on repeated sound combinations which unite the lexical components of the utterance" (in *Slovo* 116-118) and "the repetitions that connect the motto of the young princes (116) with both halves of the Grand Prince's response (117, 118)," as contrasted with the *Zadonščina*'s systematic attenuation and partial elimination of these effects. A thorough evaluation of the sound texture of the *Zadonščina* remains to be performed, and cannot be undertaken here. In general, however, we are in accord with Jakobson's conception, according to which the *Zadonščina* reveals many of the same general poetic principles as the *Slovo*, but in a much more diffuse and less highly marked way, as would be entirely consistent with the hypothesis that the *Zadonščina* is a later composition.

These remarks on the status of the *Zadonščina* lead us to comment,

finally, on the still-debated question of the *Slovo*'s authenticity, since those who deny it usually assert the author's dependence on the *Zadonščina*. If, as we have suggested, the poetics of alliteration (and related devices involving phonetic repetition) in the *Slovo* are traceable to inherited Indo-European patterns of poetic composition, and if the *Zadonščina* employs these patterns to a considerably lesser degree, then we are faced with a paradox: How is it that the alleged forger of the *Slovo*, using the *Zadonščina* as a model, reinvented, as it were, a rich and archaic poetic medium that reproduces salient features of Indo-European poetry that were not discovered until relatively recently, to say nothing of similar patterns in Old Russian prose that were not generally recognized until the middle of this century? We have, then, a situation similar to that discussed by Jakobson concerning Wackernagel's Law (1972:327-8), an important feature of ancient Indo-European syntax that indisputably both appears in the *Slovo* and is markedly absent from the *Zadonščina*: here again, Jakobson observes that "before the beginning of historical linguistics," we would have to assume that the author of the *Slovo* "divined Wackernagel's Law, having outstripped by at least a century the noteworthy discoveries facilitated by the development in the nineteenth century of the comparative-historical method in linguistics." To the extent, then, that the *Slovo*'s rich and complex technique can only be explained within the context of archaic Indo-European poetics, the analysis of the *Slovo*'s sound texture itself provides important material for answering the complex questions surrounding the history of this extraordinary poetic text.

In summary, the results of this study serve to confirm Bailey's conclusion (1971:34) that despite the relatively fixed distribution of phonemes across languages (cf. §2.2 above), "the particular phonemes chosen to occupy a given frequency can vary considerably between texts," and that "the promotion or demotion of particular phonemes" (as revealed, for example, in Bailey's prominence index or our deviation index) is "a characteristic that proves to be related. . .to the aesthetic organization of a text." More specifically, we have shown that the study of consonantal texture in the *Slovo* reveals mechanisms and relationships of great sophistication and complexity (although no doubt partly subliminal) in the work's composition. Several of the patterns discussed here have a bearing on important issues of general poetics (such as "sound symbolism"), as well as issues more specifically related to Old Russian and comparative Indo-European poetics (as in the discussion of alliteration). We have attempted to provide, then, a useful framework for the future investigation of the sound structure of the *Slovo*.

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NOTES

1. See, for instance, Cooper's recent survey of non-Soviet scholarship (1978); of the 61 bibliographical entries in his section on "Literary Analysis" (62-75), only ten are listed under a rubric "rhythmic structure," to the exclusion of any other avowedly phonetic focus. For discussions of phonetic texture, see in particular Jakobson's detailed analysis of parts of the text (1966:662-6, 680-6), as well as the more perfunctory treatments in Abicht 1907, Ržiga 1927-28, Jakubinskij 1948, and Čiževsky 1949. Most of these concern themselves with "alliteration," to which we return below.

2. For the purposes of this study, which is preliminary to a much larger work in progress, the statistical and mathematical presentation has been somewhat simplified, so as to allow for a clearer focus on issues of Slavic and general poetics.

3. Given the nature of this presentation, we use the term "deviation" in its ordinary intuitive sense, and not in its strict technical sense.

4. See further §1.2 below for more on the determination of "significance." As a final procedural point, we note that the counts just described were performed manually. As a check, however, each consonant was tabulated at least twice, and in some cases up to six times. Although a certain degree of error is inevitable, we are confident that the error in the basic count for each consonant does not exceed 2%.

5. E.g.: "(a) Les sourdes sont plus fréquentes que les sonores dans une proportion de 7 à 3 (12.77% et 5.43%); (b) Les apicales sont les plus fréquentes: environ la moitié des occlusives (9.5%) contre (3.97%) pour les labiales et (4.73%) pour les dorsales"; etc. (Guiraud 1959:103).

6. We cannot pursue here the fact that stops in the *Slovo* appear to be more frequent, in general, than in modern Russian or in the languages tabulated by Guiraud; this fact does not, in any event, vitiate the arguments concerning relative frequency just presented.

7. See Jakobson 1948; note also that when Jakobson gives two variants (with the second in parentheses), the present count is based on the first variant.

8. As for the controversy surrounding the date and authenticity of the *Slovo*: our working assumption here is that the text represents an authentic Old Russian composition of the late twelfth century; see §3.4 below for further discussion of this problem.

9. To compound the difficulty, one must also reckon, in a text of this sort, with artificially restored yers, which may function as poetic archaisms, or which may even reflect varying scribal (to say nothing of editorial) preferences.

10. Similarly Jakobson 1966:684: "In our examination of the Old Russian sound texture the difference between palatalized consonants and their non-palatalized counterparts is rarely taken into account."

11. Numbers indicate verset-numbers according to Jakobson' s reconstructed text, 1948:164-191.

12. Hereafter, numbers supplied in parentheses without the percent-sign are "deviation indexes," calculated as described in §1.1 above.

13. E.g., the following classes are only slightly less frequent than average: liquids (82.5), sonorants (87.9), fricatives (86.6), nasals (87.1); stops in general are only slightly more frequent

than average (87.8).

14. Consider, for example, the following figures: insignificantly more frequent than average: liquids (91.5), sonorants (97.5), labials (96.2), affricates (80.4); insignificantly less frequent than average: velars (99.1), nasals (99.0), stops (98.4), fricatives (85.0).

15. E.g., the following are only slightly more frequent than in the *Slovo* as a whole: liquids (98.9), stops (89.3), fricatives (87.7), nasals (97.3), labials (98.7); sonorants (98.4) and affricates (90.0) are only slightly less frequent than the norm.

16. Again, given the nature of this presentation, we forgo statistical demonstration of this point.

17. See Vine 1985.

18. In addition to the classic work of Fónagy 1963, see more recently Jakobson and Waugh 1986, and especially the important overview in Waugh 1987, with further references.

19. One of the best studies of this sort is that of Ultan 1970; for a broad range of "phonaesthetic" research, see e.g. Wescott 1980, and numerous works discussed by Waugh 1987.

20. Note also that the final alliterative sequence of voiceless labials (*pъtice . . . poli Polovьčьskomь*) is further enhanced via its enclosure of the voiced labial sequence in *biti vъ*.

21. For a recent appreciation of the considerable part played in the *Slovo* by alliteration and other forms of phonetic repetition, see the discussion by Le Guillou (1977:14-15), with further reference to treatments by Bulaxovskij and Evgen'eva.

22. The resemblance between Old Russian and Germanic alliteration would be greater only if the rhythmic structure of the *Slovo* is essentially similar to that of Old Germanic poetry; see especially Sievers' well-known analysis (1924), according to which "Die Verse sind ungefähr in dem Sinne akzentuierend gebaut wie die der altgermanischen Literaturen" (1926:6). This theory, however, remains highly controversial, at best.

23. The flippant treatment of Old Irish alliteration is particularly defective.

24. See Murphy 1961:36-9.

25. On the complex and systematic nature of alliteration in the Archaic Latin Saturnian, including structural comparisons with Germanic systems, see Valesio 1967:35-6.

26. "Eine indogermanische Stilform, vielleicht die Stilform der gemein-indogermanischen Dichtung schlechthin"; Schmitt 1967:206.

27. See also 1966:680-2, on "traditional figures of sound" in the Hypatian Chronicle and their reflection in the *Slovo*.

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